Unit 4: General Principles of Chemistry I – Rates, Equilibria and Further Organic Chemistry

Topics 4.3 – How fast? – rates	
Through practical work, students will learn about the methods used to measure reaction rates. They will collect data, analyse it and interpret the results. They can then see how a knowledge of rate equations, and other evidence, can enable chemists to propose models to describe the mechanisms of reactions.	
4.3c and 4.3d	Practical investigations in kinetics can help improve students' skills in presenting and interpretation of data. **HSW 3**
4.3d and 4.3e	This investigation can be used to bring together all the basic ideas associated with kinetics and introduce ideas of how the data can be used to help understand reactions at the molecular level. **HSW 3**
4.3h, 4.3i and 4.3j	This looks at how knowledge of rate equations, and other evidence, can enable chemists to propose models to describe the mechanisms of reactions. **HSW 1, HSW 2**
Topics 4.4 – How far? – entropy	
Topic 4.4	The study of entropy introduces students to the methods of thermodynamics and shows how chemists use formal, quantitative and abstract thinking to answer fundamental questions about the stability of chemicals and the direction of chemical change. **HSW 1**
Topics 4.5, 4.6 and 4.7 – Equilibria, application of rates and equilibrium, acid/base equilibria	
This unit tests the equilibrium law by showing the degree to which it can accurately predict changes during acid-base reactions, notably the changes to pH during titrations. *The historical development of theories explaining acids and bases shows how scientific ideas change as a result of new evidence and fresh thinking.*	
4.5c	Students could be encouraged to analyse data from theory or experiments to establish equilibrium relationships. They should appreciate that the relationships are empirical and apply only to the system under consideration. **HSW 1, HSW 5**
4.5f and 4.5h	These sections link thermodynamic theories with equilibrium situations and allows students to obtain a deeper understanding of the reasons for observed effects of changes of conditions on equilibrium. **HSW 1**
4.6a to 4.6d	This topic applies the material in topics 4.3, 4.4 and 4.5 of this unit to industrial situations and begins to look at the way industry can use models/theories, including kinetics, equilibrium and thermodynamic data, in the search for better yields at a lower cost to the environment. **HSW 1, HSW 2, HSW 8, HSW 9, HSW 10, HSW 12**

4.7a to 4.7c	Students could look at how our understanding of acid and bases has developed over the last 150 years as knowledge has increased and new theories of atomic behaviour have been formulated. **HSW 1**
4.7e	Students can use the model of ionic dissociation to explain acid/base behaviour. **HSW 1**
4.7g	Determination of Ka from pH and concentration data provides a good example of when it is appropriate to make approximations in chemistry and when it is not. **HSW 2**
4.7m	This can be used as an example of how chemists can help society to understand and minimise risk. **HSW 9**
Topics 4.8 – Further organic chemistry	
4.8.1d	Analysis of data for optical activity — i.e. production of a racemic mixture or a single enantiomorph could be used to provide evidence for proposed mechanisms. **HSW 3, HSW 5**
Topics 4.9 – Spectroscopy and chromatography	
Topic 4.9	This considers how chemists use the interaction of various types of radiation with molecules, including the use of: ■ IR by the chemical industry as a non-invasive technique to follow the extent of a reaction ■ microwaves to provide a heat source in a closed system that can allow reactions to proceed at a temperature double that of the highest boiling point component thus reducing the time for a reaction by orders of magnitude, saving time and energy ■ nmr spectroscopy in medicine and industry. It also includes the use of HPLC as a method of rapid separation of liquid mixtures prior to further analysis. **HSW 9, HSW 10, HSW 11, HSW 12**